Synthetic Genomics Data Generation and Evaluation for the Use Case of Benchmarking Somatic Variant Calling Algorithms

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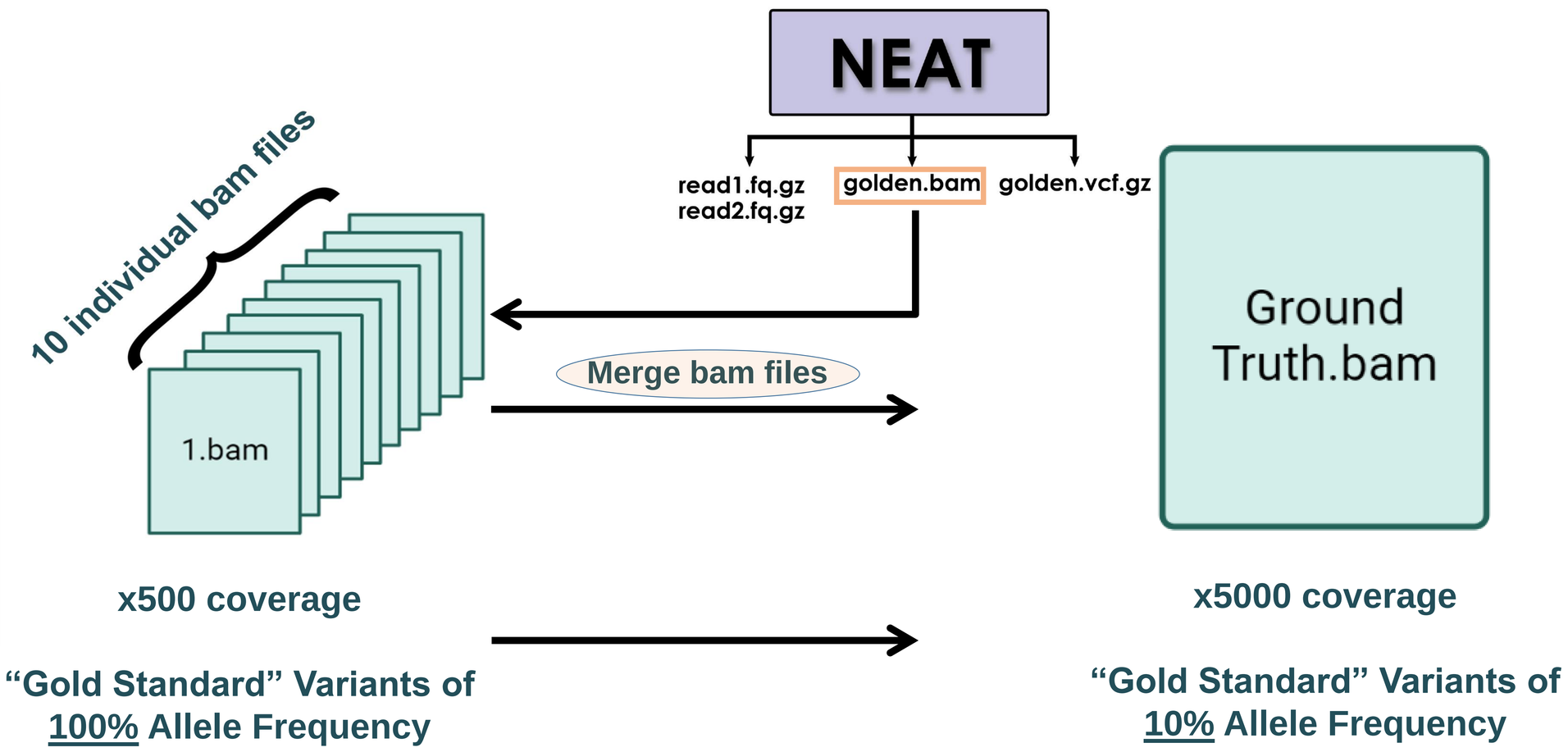
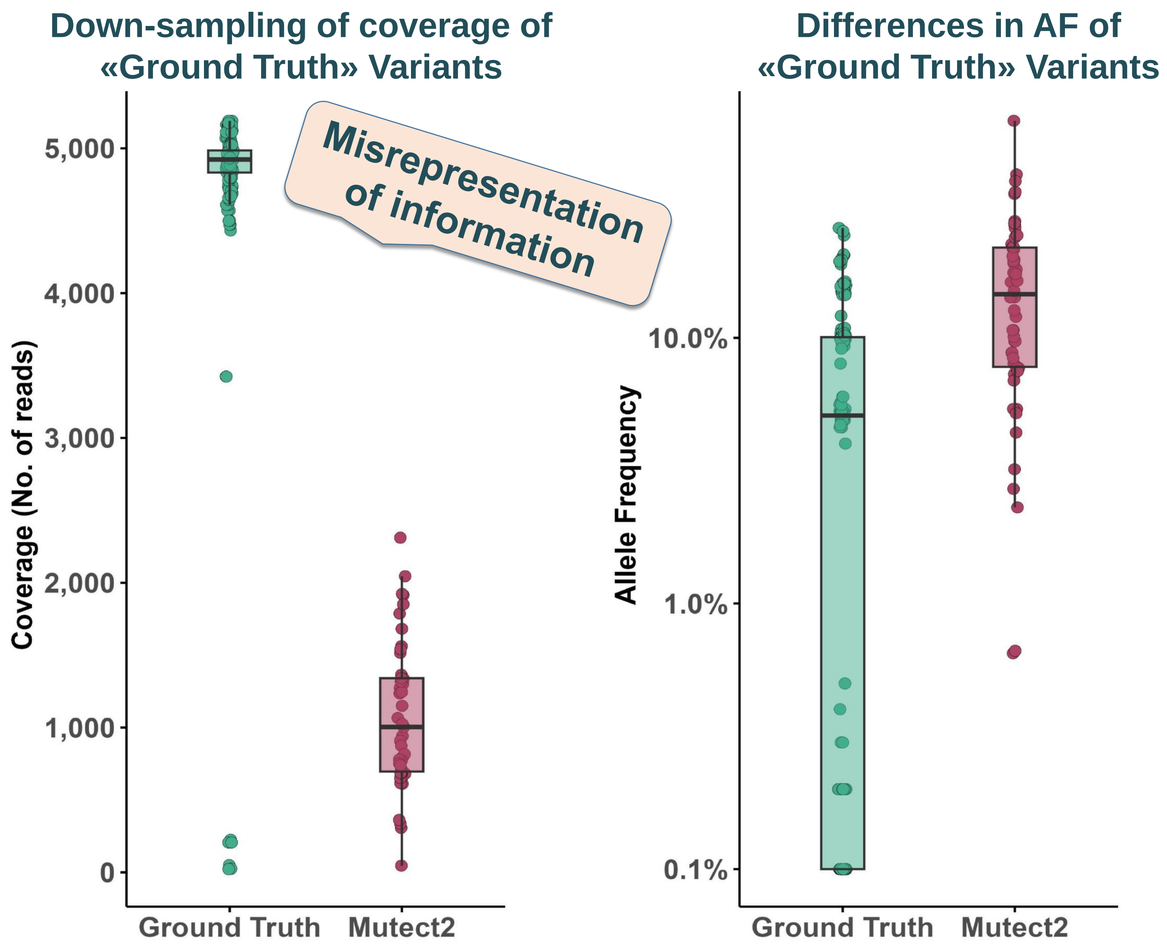
1 Synthetic «Ground Truth» Dataset Generation

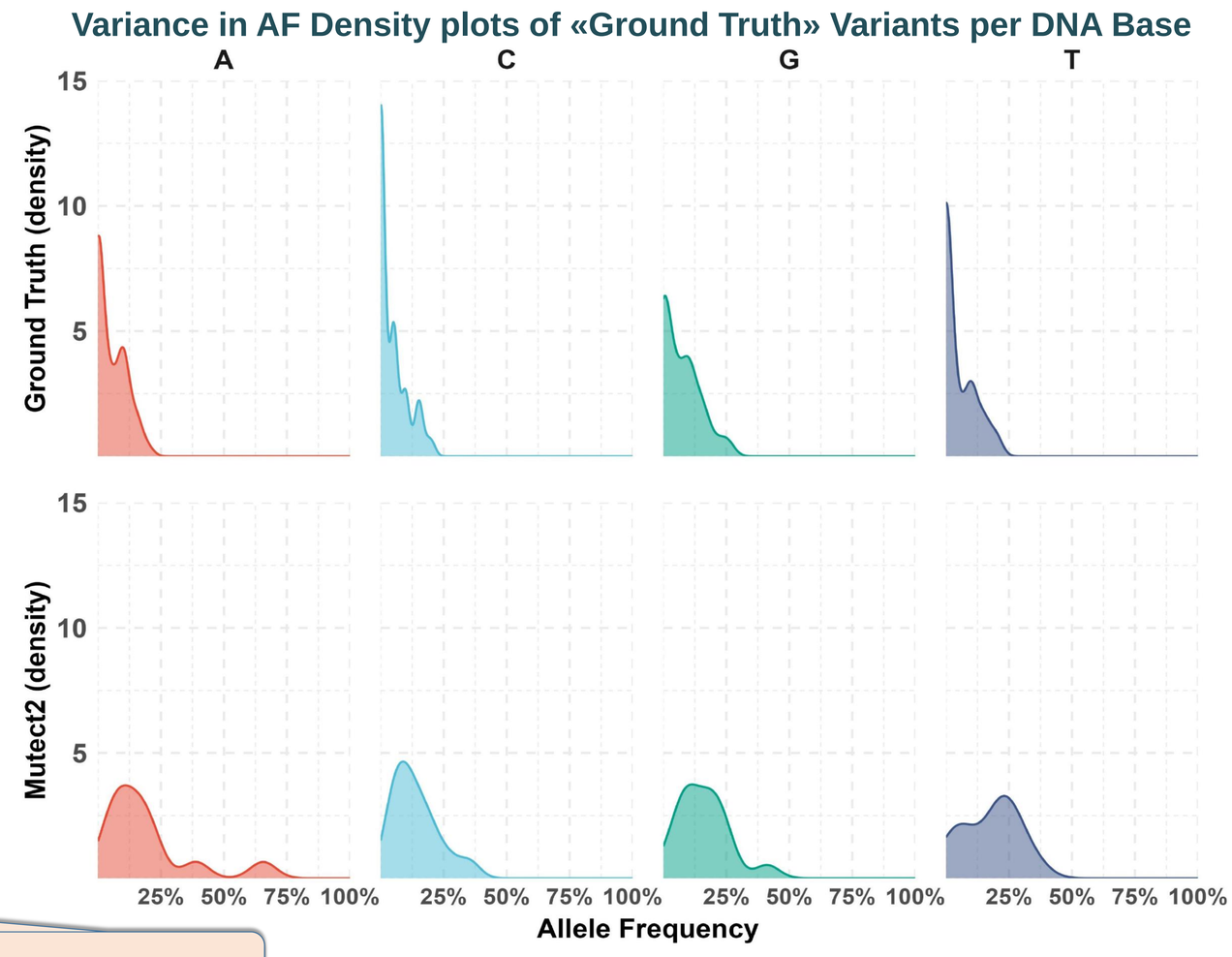
Highlights

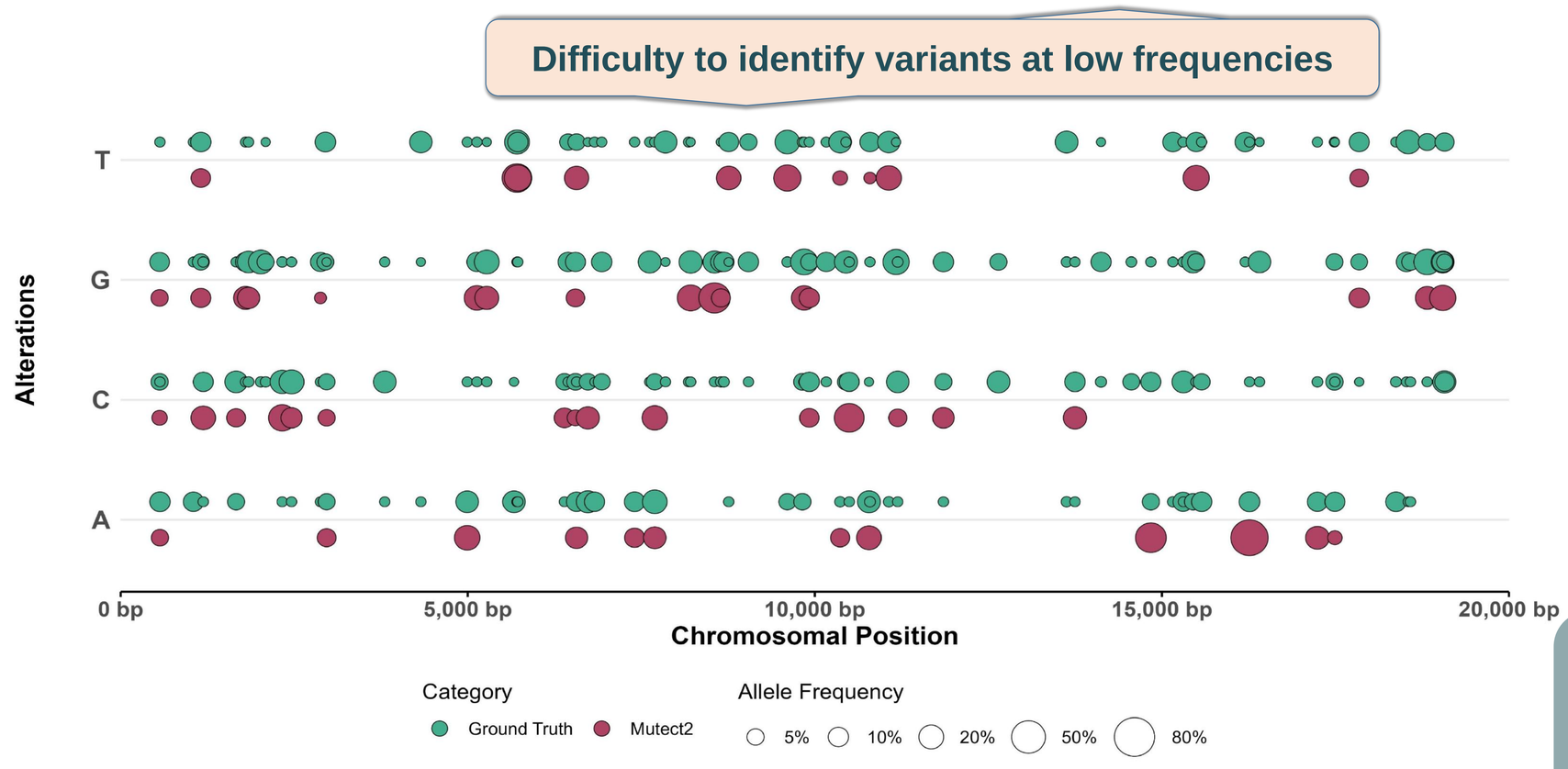
- Generation of **synthetic genomics** data based on *TP53* gene
- Define **«Ground Truth»** SNIPs and INDELs in order to **benchmark** somatic variant callers
- Investigate the impact of variant callers in variants at **low frequencies**



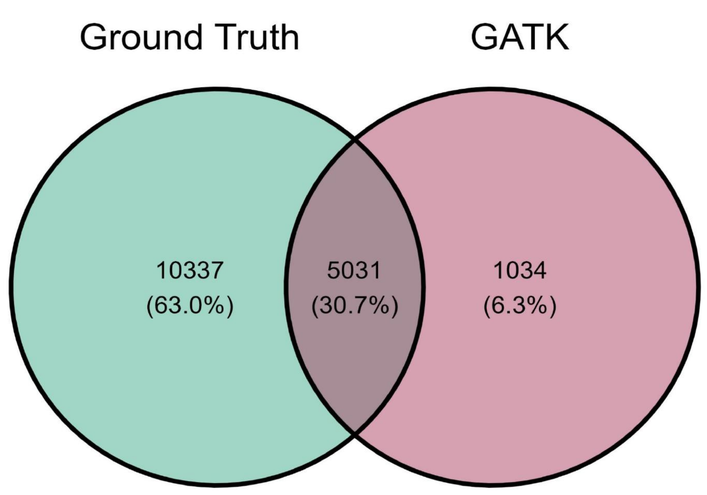
2 Benchmarking GATK-Mutect2







Divergence in the identification of SNIPs and their AF of «Ground Truth» Variants



Venn plot of the Overall Variants

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